Risk Alert Classifier: Final Report

Predicting High-Risk customers (Risk Status = 1) from demographic, transaction and credit-behaviour data. Dataset: 4,600 customers, 17 input features, 12.1% High Risk.

1. Executive summary

- **Recommended model: Tuned Logistic Regression** (L1 penalty, C = 0.3, class_weight = balanced, threshold 0.50). On the held-out test set (920 customers, 111 high risk) it produced **0 false negatives and 2 false positives** (recall 1.000, precision 0.982, AUC 1.000).
- Repeated cross-validation on the training data gives recall 0.988 +/- 0.012 and precision 0.961 for this model, the best recall among models that meet a 0.95 precision floor.
- Accuracy is misleading here: predicting "Low Risk" for everyone scores 87.9% with zero recall.
- Balancing techniques (under/over-sampling, SMOTE, ADASYN) did not help Logistic Regression, but raised recall for the Decision Tree (0.838 to about 0.93) and rescued the Random Forest under severe imbalance. Each trades precision for recall.
- The near-perfect scores indicate a very clean, probably synthetic dataset; real-world performance should be expected to be lower.

2. Approach

Step	What was done
Preparation	Dropped customer_id; converted last_transaction_date to days since last transaction; stratified 80/20 split (High-Risk share 12.1% in train and test); one-hot encoding; standardisation; multivariate KNN imputation (k = 5) of 7 columns with 2-5% missing values, all fitted on the training set only.
Baseline	Logistic Regression on the imbalanced data.
Imbalance	Random under-sampling, random over-sampling, SMOTE and ADASYN applied to the training set only; compared on Logistic Regression, Decision Tree and Random Forest, plus a severe-imbalance stress test (1.8% High Risk).
Tree models	Decision Tree (overfitting analysis) and Random Forest.
Tuning	Randomized Search CV (Decision Tree, Random Forest), Grid Search CV (Random Forest and Logistic Regression); 5-fold stratified CV with F2 scoring so that recall counts more than precision.
Selection	ROC/AUC for all models, repeated 5x2 CV on the training set, rule: highest CV recall with CV precision >= 0.95.

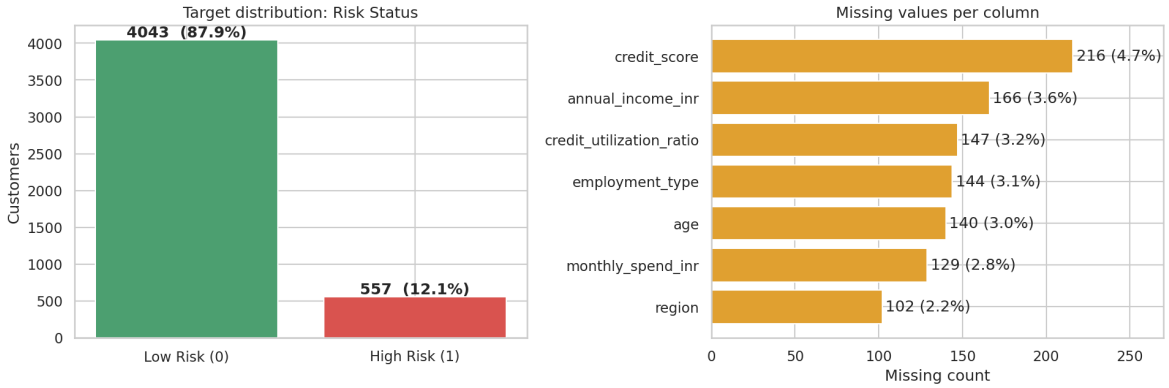


Figure 1. Target distribution and missing values.

3. Best classification model and justification

Model (test set)	Recall	Precision	F1	AUC	FN	FP
Logistic Regression (baseline)	1.000	1.000	1.000	1.0000	0	0
Tuned Logistic Regression (Grid)	1.000	0.982	0.991	1.0000	0	2
Tuned Random Forest (Grid)	1.000	0.974	0.987	1.0000	0	3
Tuned Random Forest (Randomized)	1.000	0.974	0.987	1.0000	0	3
Random Forest (default)	0.982	1.000	0.991	0.9999	2	0
Tuned Decision Tree (Randomized)	0.973	0.783	0.868	0.9832	3	30
Decision Tree (default)	0.838	0.903	0.869	0.9127	18	10

Model (5x2 repeated CV, training set)	Recall	Precision	F2	AUC
Logistic Regression (baseline)	0.975 +/- 0.018	0.989	0.978	0.9995
Tuned Logistic Regression (Grid)	0.988 +/- 0.012	0.961	0.982	0.9995
Decision Tree (default)	0.881 +/- 0.040	0.885	0.881	0.9326
Tuned Decision Tree (Randomized)	0.951 +/- 0.022	0.755	0.903	0.9752
Random Forest (default)	0.930 +/- 0.038	0.987	0.941	0.9965
Tuned Random Forest (Randomized)	0.964 +/- 0.022	0.965	0.964	0.9977
Tuned Random Forest (Grid)	0.966 +/- 0.020	0.961	0.965	0.9975

The test set contains only 111 high-risk customers, so one error changes recall by about 0.9 points; the cross-validated comparison is therefore used for the final decision. A linear model wins because the risk label is close to a weighted combination of the credit-behaviour features, so extra flexibility adds variance but no signal. It is also the most explainable option: income (negative) and late-payment days, utilisation, debt balance and missed payments (positive) are its largest drivers.

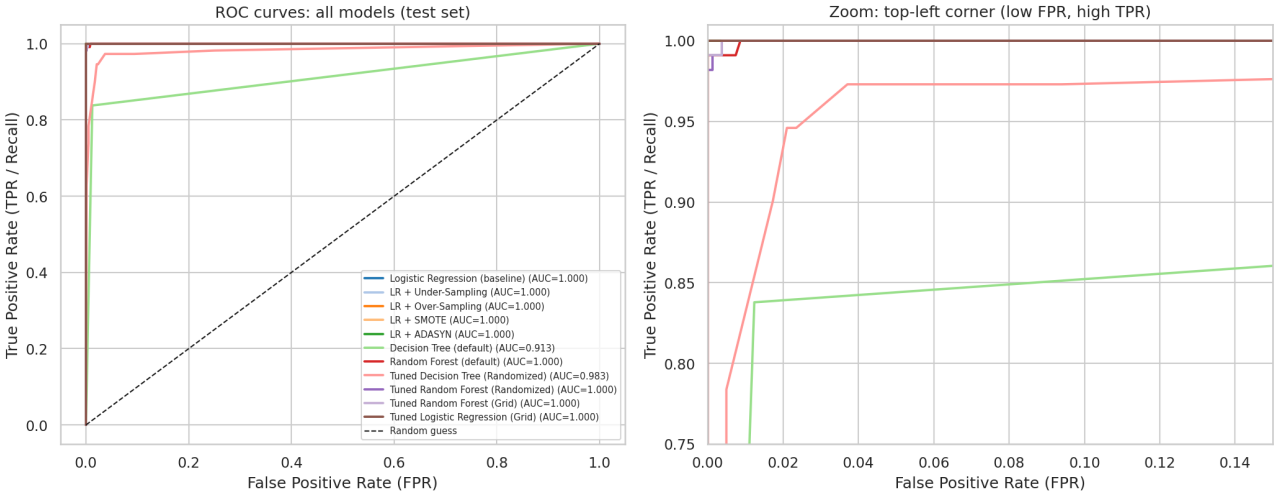


Figure 2. ROC curves for all models (right: zoom on the top-left corner).

4. Impact of imbalance-handling techniques

Base model	Balancing	Precision	Recall	F1	FN	FP
Decision Tree	nan	0.903	0.838	0.869	18	10
Decision Tree	Under-Sampling	0.693	0.937	0.797	7	46
Decision Tree	Over-Sampling	0.911	0.928	0.920	8	10

Base model	Balancing	Precision	Recall	F1	FN	FP
Decision Tree	SMOTE	0.832	0.937	0.881	7	21
Decision Tree	ADASYN	0.811	0.928	0.866	8	24
Random Forest	nan	1.000	0.982	0.991	2	0
Random Forest	Under-Sampling	0.910	1.000	0.953	0	11
Random Forest	Over-Sampling	0.991	0.964	0.977	4	1
Random Forest	SMOTE	0.991	0.982	0.986	2	1
Random Forest	ADASYN	0.973	0.973	0.973	3	3

Table: effect of balancing on tree models (test set). For Logistic Regression, recall stayed at 1.000 with 1-2 extra false positives under every technique.

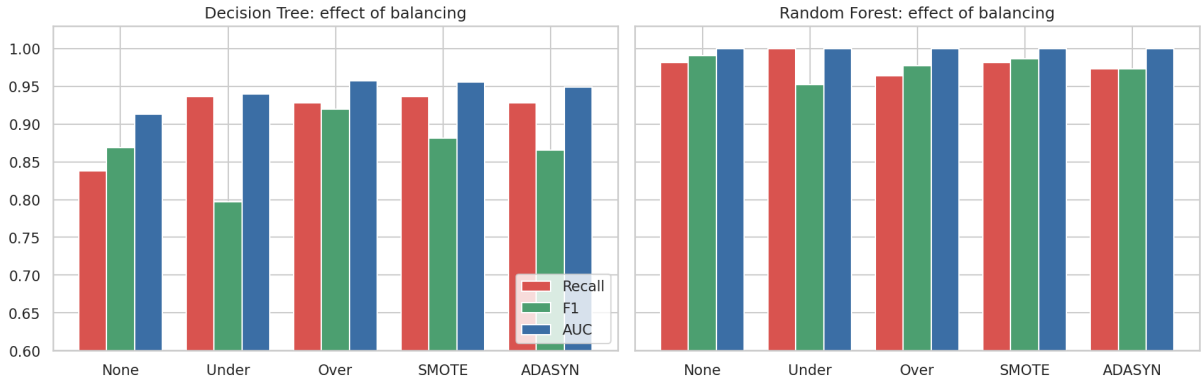


Figure 3. Recall, F1 and AUC for the tree models under each balancing technique.

Stress test: training set reduced to 60 high-risk customers (1.8%).

Model	Balancing	Precision	Recall	F1	FN	FP
Logistic Regression	nan	1.000	0.991	0.996	1	0
Random Forest	nan	1.000	0.793	0.884	23	0
Logistic Regression	Under-Sampling	0.867	1.000	0.929	0	17
Random Forest	Under-Sampling	0.895	1.000	0.945	0	13
Logistic Regression	Over-Sampling	0.982	0.991	0.987	1	2
Random Forest	Over-Sampling	1.000	0.712	0.832	32	0
Logistic Regression	SMOTE	0.991	0.991	0.991	1	1
Random Forest	SMOTE	1.000	0.856	0.922	16	0
Logistic Regression	ADASYN	0.982	0.991	0.987	1	2
Random Forest	ADASYN	1.000	0.856	0.922	16	0

Overfitting check: the unpruned Decision Tree scores 1.000 F1 on training data but 0.869 on test data (gap 0.131); the Random Forest's gap is 0.009.

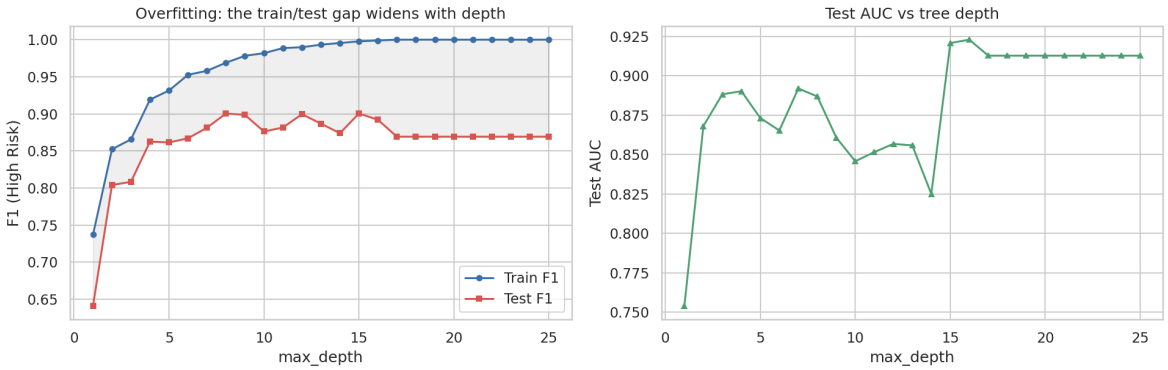


Figure 4. Train vs test F1 as tree depth grows.

5. Business interpretation of false positives and false negatives

Error	Meaning	Business impact	Final model (test)
False negative (Type-II)	High-risk customer predicted Low Risk	Undetected default / fraud exposure: the costly error, so minimised first	0 of 111
False positive (Type-I)	Low-risk customer flagged High Risk	Analyst review time, possible customer friction	2 of 809

Threshold analysis on out-of-fold predictions shows the default 0.50 cut-off is already F2-optimal for the class-weighted model. A lower cut-off (about 0.32) is available as a high-recall option: on the test set it caught no additional customers but doubled the false positives.

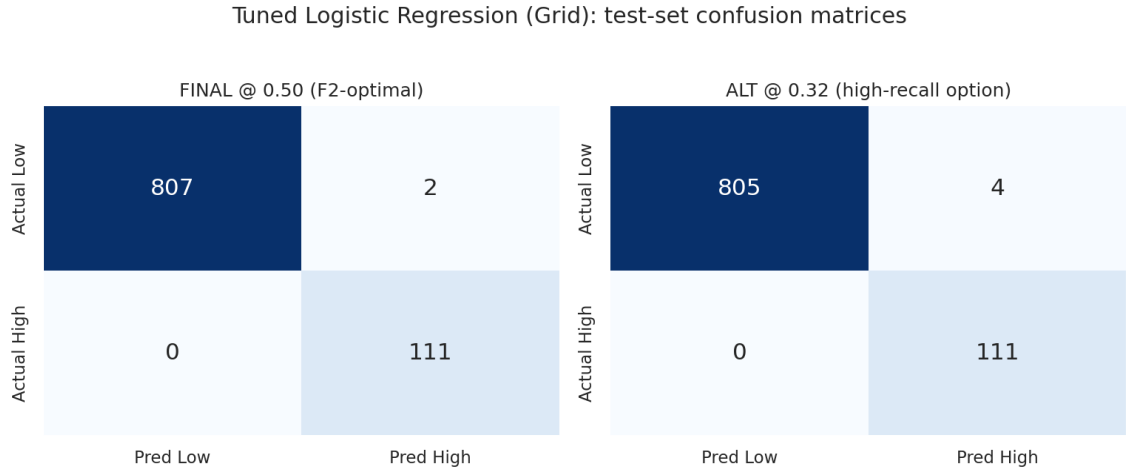


Figure 5. Final model confusion matrices on the test set.

6. Conclusions and recommendations

- Deploy the tuned Logistic Regression (threshold 0.50) as the alerting model, with human review of each alert.
- Keep the lower-threshold (high-recall) option available if the business wants a bigger safety margin.
- When tree ensembles are used, apply class weights or under-sampling/SMOTE and report recall and precision, not accuracy.
- Monitor recall on confirmed outcomes and drift in late-payment days, utilisation and income; retrain regularly.

- Run a fairness review before real use (the Other gender group has only 43 customers) and validate on fresh, out-of-time data.

7. Limitations

- Very high scores and the absence of leakage point to a clean, probably synthetic dataset; expect lower real-world performance.
- Only 111 high-risk customers in the test set; differences of one or two customers are within noise.
- SMOTE and ADASYN results in this report come from built-in implementations of the same algorithms because imbalanced-learn was unavailable when outputs were generated; the notebook uses the library automatically when installed.